ISYS 1108 / 3474 Software Engineering Project Management

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Week 3: Lectorial

What's next...

RMIT UNIVERSITY

Acknowledgement of Country

RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceded lands we conduct the business of the University.

RMIT University respectfully acknowledges their Ancestors and Elders, past and present.

RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waters across Australia where we conduct our business.

Classes and Deadlines

Mid-semester break is between Weeks 5 and 6

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Lectorials												
Workshops						Team-based assignment activities						
A1: SE Challenge				Friday: Challenge 1 10%					Friday: Challenge 2 10%		Friday: Challenge 3 10%	
A2: Team-based assignment								Due on Friday: 30%				
A3: Individual Assignment												Due on Friday: 40%

Our plan for this session

- 😊 SE Challenge 1: next Friday (Week 4)
- 😊 User Story Cards and Acceptance Criteria
- 😊 Recap quiz: agile vs Scrum, Scrum roles and artefacts
- 😊 DoD vs Acceptance Criteria
- 😊 … more exercises on User Story Cards

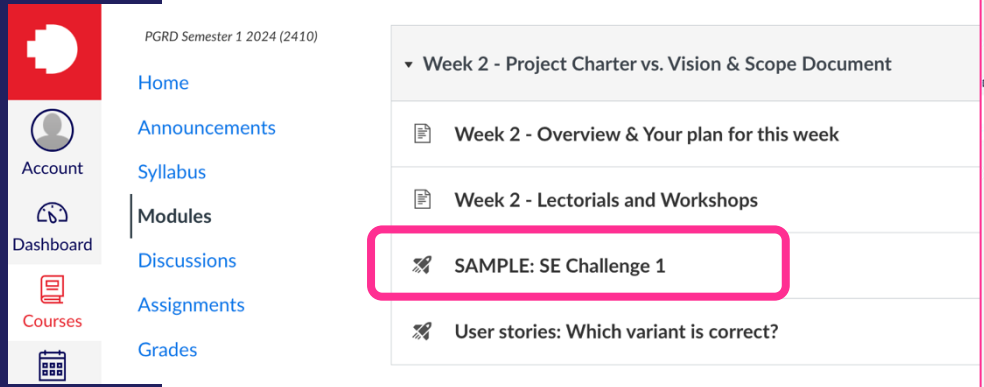
+ any questions you have on related material ☺

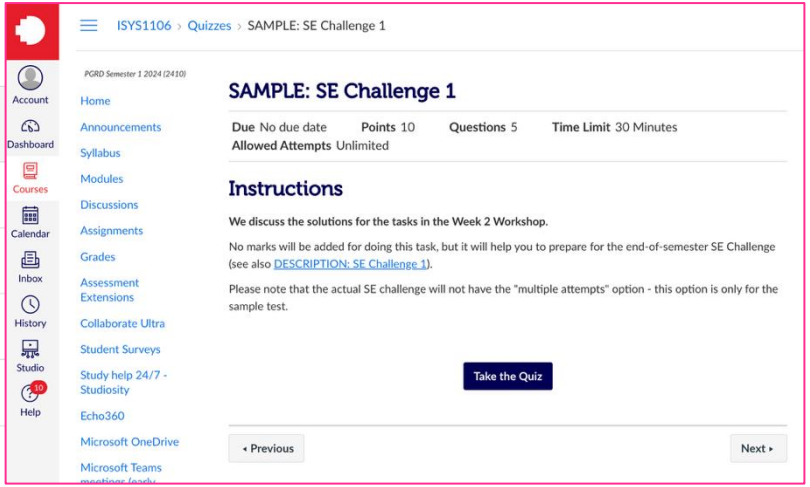
SE Challenge: Friday of Week 4

10% of the final grade

This challenge will be provided as a Canvas quiz, available to you within **24h, from 12:00am till 11:59pm, on the corresponding date**.

You will be examined on the course material presented and discussed **on Weeks 1 – 3**, especially focusing on the following topics: **Work Breakdown Structure, Vision & Scope document, Project Charter**.





SE Challenge: 27 March (Friday of Week 4)

You will have ONE attempt at this assessment.

Once you open the test you must complete it in the allocated time. **To complete this task, you will have exactly 30 minutes from the time you start the quiz (limited by the due time on 11:59pm).**

If your test is disrupted due to: PC crash, internet dropout, etc., please take a screenshot or use your phone to take a photo as evidence, let the course coordinator know and apply for a special consideration. Please note that the course coordinator cannot provide any extensions for this assessment, only spec. cons. office can approve a deferred challenge.

User Story Cards

User Story vs. User Story Card

User Story Card (USC)
=

User Story (US)

+ unique ID

+ Priority,

+ Effort,

+ Acceptance criteria

Why cards?

User stories emerged in Extreme Programming (XP)

The early XP literature talks about **story cards** rather than **user stories**.

Why?

User stories were captured on paper cards.

Benefits:

1. Paper cards are cheap and easy to use.
2. They facilitate collaboration: Every one can take a card and write down an idea.
3. Cards can be easily grouped on the table or wall to check for consistency and completeness and to visualize dependencies.

Even if your stories are stored electronically, it might be worthwhile to use paper cards when you write new stories.

https://www.romanpichler.com/blog/10-tips-writing-good-user-stories/

Acceptance criteria

Condition of satisfaction: a high-level acceptance test (**acceptance criteria**) that will be true after the user story is complete

US1: **As a** Premium member, **I want** to be able to cancel my hotel reservations at any time **so that** I can save on cancellation penalty.

IDs required to achieve traceability

AC1.1: Verify that a Premium member can cancel without any penalty.

AC1.2: Verify that the hotel is notified of cancellation request.

AC1.3: Verify that the Premium member receives a confirmation that their cancellation request is complete, and no penalty has been charged.

This is the simplest way to specify ACs, but it has **many** disadvantages…

Acceptance criteria (AC): Given – When –Then

Behavior-driven development

There should be only one action in the WHEN-part
→ If you need to cover 2 or more actions, you will need to specify 2 or more ACs

GIVEN some context

WHEN some action is carried out

THEN a particular set of measurable consequences should be observable

These ACs look more complex, but most companies use this format as it's more precise.

Specific

Measurable

Attainable

Realistic

Timely

Acceptance criteria: Let’s convert Verify-style in Given-When-Then

AC1.1: Verify that a Premium member can cancel without any penalty.

How would you reformulate this AC?

Acceptance criteria: Given-When-Then format AC1.1

AC1.1: Verify that a Premium member can cancel without any penalty.

→

AC1.1:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then this reservation is cancelled without without any penalty.

This is one of possible ways to formulate this criterion in Given-When-Then format

Acceptance criteria: Given-When-Then format AC1.1

AC1.1: Verify that a Premium member can cancel without any penalty.

→

AC1.1:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then the system cancels that reservation without without any penalty.

This is one of possible ways to formulate this criterion in Given-When-Then format

Acceptance criteria: Let’s convert Verify-style in Given-When-Then

AC1.2: Verify that the hotel is notified of cancellation request.

How would you reformulate this AC?

Acceptance criteria: Given-When-Then format AC1.2

AC1.2: Verify that the hotel is notified of cancellation request.

→

AC1.2:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then the hotel is notified of cancellation request.

This is one of possible ways to formulate this criterion in Given-When-Then format

Acceptance criteria: Given-When-Then format AC1.2

AC1.2: Verify that the hotel is notified of cancellation request.

→

AC1.2:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then the system sends an email to the hotel to notify regarding cancellation request.

This is one of possible ways to formulate this criterion in Given-When-Then format

Acceptance criteria: Let’s convert Verify-style in Given-When-Then

AC1.3: Verify that the Premium member receives a confirmation that their cancellation request is complete, and no penalty has been charged.

How would you reformulate this AC?

Acceptance criteria: Given-When-Then format AC1.3

AC1.3: Verify that the Premium member receives a confirmation that their cancellation request is complete, and no penalty has been charged.

→

AC1.3:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then the user receives a confirmation that their cancellation request is complete without any penalty.

This is one of possible ways to formulate this criterion in Given-When-Then format

Acceptance criteria: Given-When-Then format AC1.3

AC1.3: Verify that the Premium member receives a confirmation that their cancellation request is complete, and no penalty has been charged.

→

AC1.3:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then the system sends to the user an email with a confirmation that their cancellation request is complete without any penalty.

This is one of possible ways to formulate this criterion in Given-When-Then format

Acceptance criteria: Given-When-Then format – alternative solution

AC1.1: Verify that a Premium member can cancel without any penalty.

AC1.2: Verify that the hotel is notified of cancellation request.

AC1.3: Verify that the Premium member receives a confirmation that their cancellation request is complete, and no penalty has been charged.

→

AC1.1:

Given a user is logged in as a Premium member AND the user has a hotel reservation

When the user clicks on “cancel reservation” button associated with that reservation

Then the system cancels that reservation without without any penalty AND the system sends an email to the hotel to notify regarding cancellation request AND the system sends to the user an email with a confirmation that their cancellation request is complete without any penalty.

To wrote ACs, a User Story should be small

Larger stories = hidden assumptions and ambiguity during the development

Example: As a website user, I want to post my resume so that I can be contacted regarding job interviews.

Too vague and compound story because

- A resume includes separate sections for education, prior jobs, salary history, publications, etc.
- Users can mark resumes as inactive
- Users can have multiple resumes
- Users can edit resumes
- Users can delete resumes

What to do with a bigger story?

Big, coarse-grained, sketchy

Small, detailed, specific

Epics

https://www.ibm.com/developerworks/rational/library/user-stories-product-backlog/index.html

- A large user story is typically referred to as an **epic**
- There is no “magic size” at which we start calling a user story an epic
- Generally: an **epic** is a user story that will take more than **one** or two sprints to develop and test
- Epics are split into smaller user stories, so that a story can be completed within a sprint

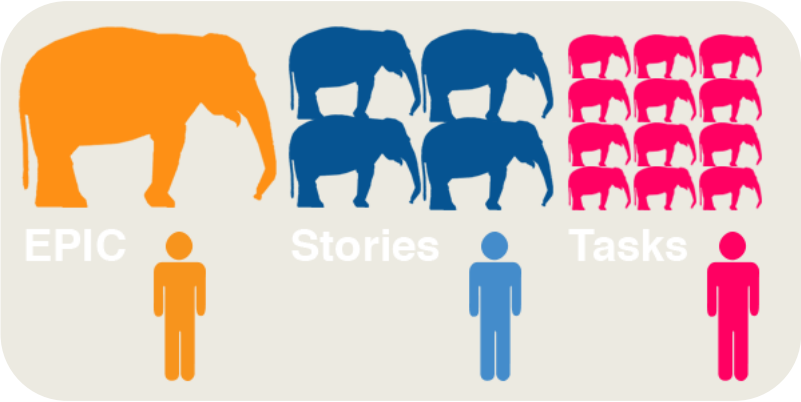
Is this an epic?

As a user, I want to log in to the system so that my information can be accessed only by me.

Assume that the product owner clarifies that this simple story is intended to cover everything related to logging in:

- requesting a new password,
- changing the password,
- ...

It is about more than pressing a Log in button on one screen.



From an epic to a user story

As a user, I want to log in to the system so that my information can be accessed only by me.

- As a new user, I want to register by creating a username and password so that the system can remember my personal information.
- As a registered user, I want to log in with my username and password so that I can trust the system.
- As a registered user, I want to change my password so that I can keep it secure or make it easier to remember.
- As a registered user, I want the system to warn me if my password is easy to guess so that my account is harder to break into.
- As a registered user, I want to be able to request a new password so that I am not permanently locked out if I forget it.
- As a registered user, I do not want to be sure if it was the username, password, or both that was wrong when my login attempt fails so that someone trying to impersonate me will have a harder time doing so.
- …

Epics:

- groups of related stories
- comprise a complete work flow for a user
- the stories that comprise an epic may be completed independently, but their business value isn't realised until the entire epic is complete
- it rarely makes sense to deliver an epic until all of the underlying stories are complete

Themes:

- groups of related stories
- contribute to a common goal or are related in some obvious way, such as all focusing on a single customer.
- stories in a theme do not need to encapsulate a specific work flow or be delivered together.
- some stories in a theme may be dependent on one another, but are independent enough to be delivered separately and still provide some business value

What makes a product backlog good / bad?

The product backlog should be available for anyone to see.

The backlog must be the single source of requests.
It's important for the source of requests to be centralized because the entire list needs to be ordered.

The product backlog is a living artifact and is never complete.
It's very normal to identify new items, remove items that are no longer considered valuable and to morph existing items.
The product backlog lives for as long as the product lives.

Using the backlog as a dumping ground of ideas to appease stakeholders.

Having more than 1 backlog

The Scrum guide suggests you spend no more than 10% of your team's time keeping the backlog in working order.

Grooming your backlog ensures that it always exhibits the healthy characteristics that make the backlog a pleasure to work with. If you don't groom the backlog you'll run into difficulty planning and delivering backlog items.

Is our Product Backlog correct?

- ✓ Are the PBI IDs unique? Do the IDs make sense?
- ✓ Are the PBI written as reasonable user stories? (reasonable semantics, correct format As a… I want to … so that …)
- ✓ Is the product backlog sorted by the priorities of the PBIs?
- ✓ Are the priorities correct wrt. notes made during the initial requirements elicitation interviews with the stakeholders?
- ✓ Are the Owners reasonable wrt. the PBIs?
- ✓ Are the Efforts reasonable wrt. the PBIs?
- ✓ Are the DoDs reasonable wrt. the PBIs?

… and we also need to check COMPLETENESS of the Product Backlog (i.e. whether all needed requirements are covered)

PGRD Semester 1 2024 (2410)

Account
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Modules
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Assignments
Grades
Assessment

Week 3 - Agile/Scrum: User story cards, Acceptance criteria

Week 3 - Overview & Your plan for this week

WIL Agreement - Mandatory
Mar 21 0 pts

Week 3 - Lectorials and Workshops

Recap: Agile vs Scrum, Scrum roles and artefacts

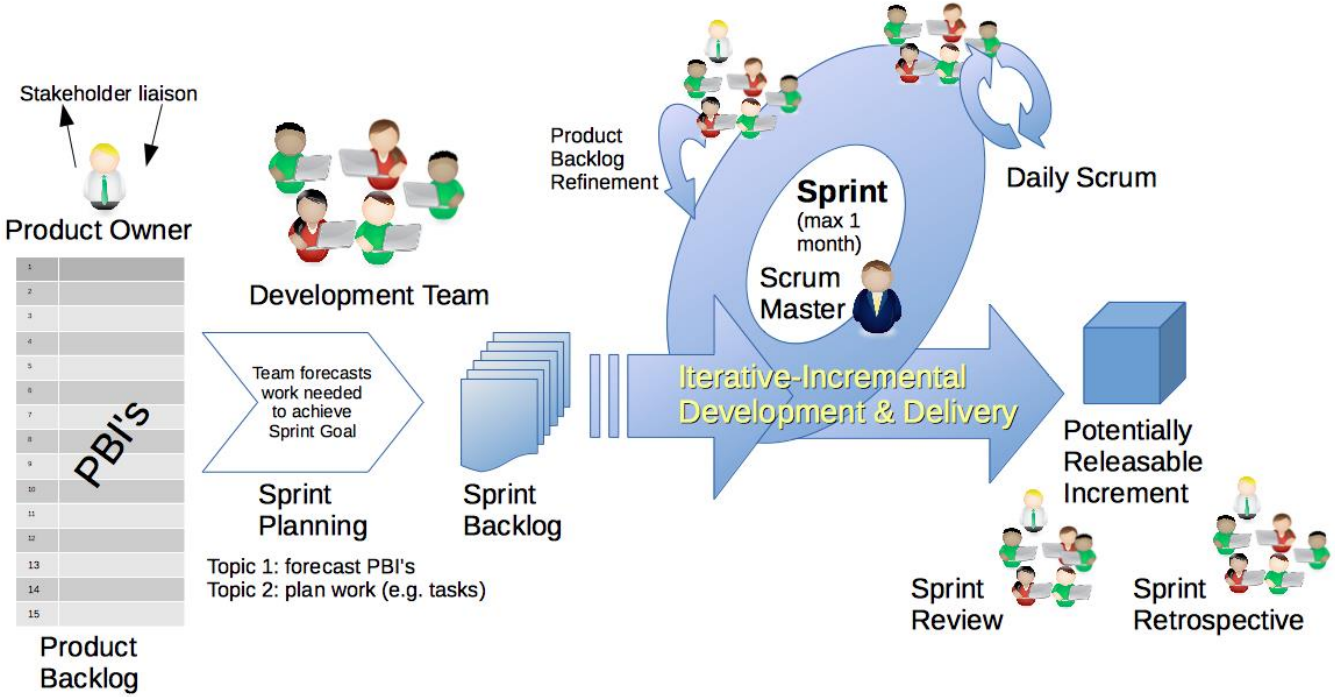
Recap: Agile vs Scrum, Scrum roles and artefacts

Due No due date **Points** 10 **Questions** 10
Time Limit 10 Minutes **Allowed Attempts** Unlimited

Instructions

This is ungraded, unlimited attempt quiz for you to self-check your knowledge of Scrum roles and artefacts.

Take the Quiz


Stakeholder liaison
Product Owner
Development Team
Product Backlog Refinement
Sprint
(max 1 month)
Scrum Master
Daily Scrum
PBI'S
Team forecasts work needed to achieve Sprint Goal
Sprint Planning
Sprint Backlog
Iterative-Incremental Development & Delivery
Potentially Releasable Increment
Topic 1: forecast PBI's
Topic 2: plan work (e.g. tasks)
Sprint Review
Sprint Retrospective
Product Backlog

Definition of Done

DoD

DoD vs. Acceptance Criteria

DoD is intended to be universally applicable

(quality!)

Acceptance Criteria are specific for each feature

(functionality!)

DoD should NOT include any functionality!

Definition of Done (DoD)

- The exit-criteria to determine whether a product backlog item is complete.
- In many cases the DoD requires that all regression tests should be successful.
- The definition of "done" may vary from one Scrum team to another, but must be consistent within one team.

- **List of activities that add verifiable/demonstrable value to the product**
- DoD is **orthogonal to user acceptance criteria** (functional acceptance) for a feature: It is a comprehensive **checklist** of necessary, value-added activities that assert the **quality of a feature and not the functionality** of that feature.
- Ensures that only truly done features are delivered, not only in terms of functionality but in terms of quality as well.

Definition of Done

Goals:

- to build a common understanding within the Team about Quality and Completeness
- to be a checklist that User Stories (or PBIs) are checked against
- to ensure the increment shipped at the end of the Sprint has high quality and that the quality is well understood by all involved.

Example of DoD:

- ✓ Code is either reviewed or produced with a pair-programming method
- ✓ The code is covered by Unit Tests (minimum of 70% code coverage) and all tests are Green (*means it has passed*)
- ✓ Automated acceptance tests (based on the Acceptance Criteria of the User Story) are conducted and passed
- ✓ Integration tests of the affected areas are conducted and passed
- ✓ Usability Review Completed, Tested on iPhone 6-11
- ✓ Regression testing

Definition of Done (DoD, Done Done)

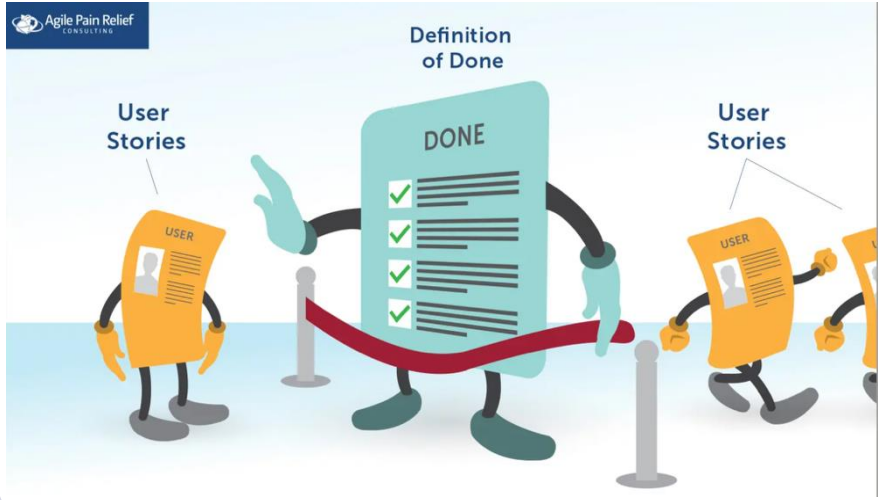
- should focus on how you will ensure the quality of the product you develop – what kind of tests you are going to conduct, what quality criteria you will apply to each item
- DoD == the rules that the development team, Product Owner, and to some extent the organisation have agreed on when they mean a story is ready for delivery

DoD vs. Acceptance Criteria

DoD is intended to be universally applicable (quality!)

Acceptance Criteria are specific for each feature (functionality!)

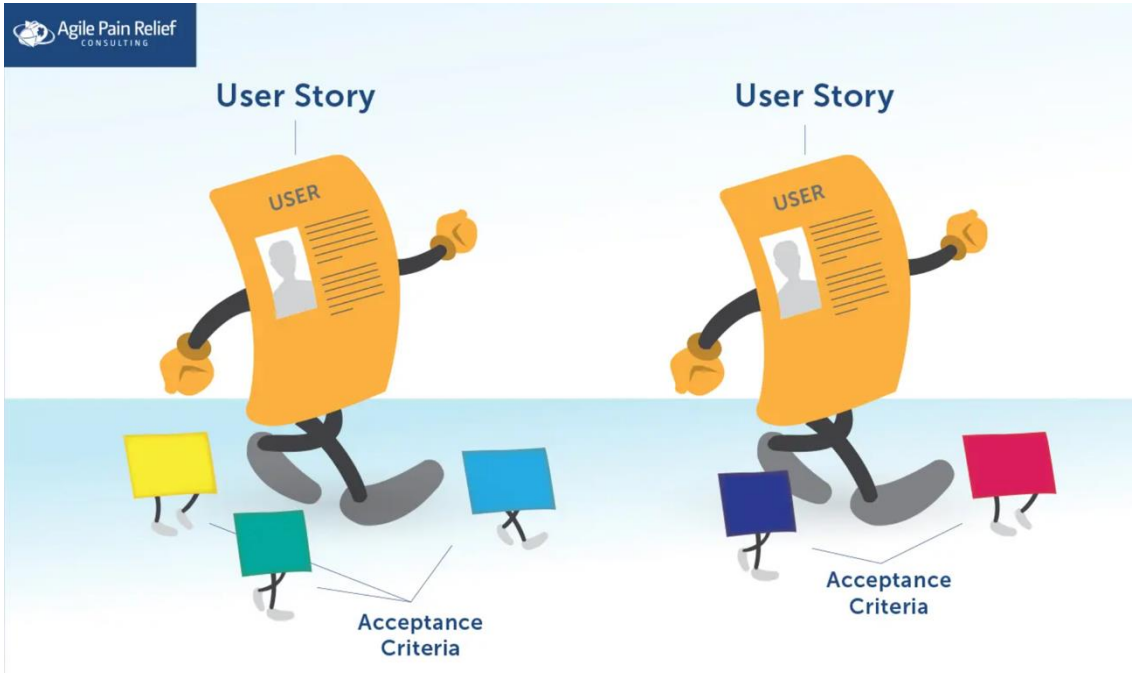
DoD should NOT include any functionality!



Example of DoD

- Unit testing completed (coverage 80%)
- Peer reviewed
- Architecture updated (if necessary)
- Functional testing based on ACs completed
- Regression testing completed
- Performance testing benchmark achieved
- User documentation updated to reflect new story
- Approved by the Product Owner

User stories and Acceptance criteria



DoD: Levels

There might be different DoD at various levels:

- DoD for a **Product Backlog Item** (e.g., writing code, tests and all necessary documentation)
- DoD for a **Sprint** (e.g., install demo system for review)
- DoD for a **Release** (e.g., writing release notes)

  - Can we do this activity for each feature? If not, then
  - Can we do this activity for each sprint? If not, then
  - We have to do this activity for our release!

You also might read this:
http://www.scrumalliance.org/community/articles/2008/september/what-is-definition-of-done-(dod)

Let’s do more exercises on USCs

Story # 5		Priority	Medium
		Effort	8
As a	New franchisee user		
I want to	create a franchisee account		
So that	I can start working with Victoria Farms		
Acceptance criteria	**AC 5.1** **Given** a new user is on the home page of the system and isn't logged in **When** the user clicks the icon "Add a franchisee" **Then** the user is forwarded to the page "Add new franchisee" **AC 5.2** **Given** the user is on the "Add new franchisee" page AND the login details of a new franchisee are entered correctly (the franchisee email is not recorded in the system) **When** the user clicks on the "Add" button **Then** a message is displayed "Franchisee is added to the system." AND the new login details are added to the system AND the system sends a notification email to franchisee **AC 5.3** **Given** the user is on the "Add new franchisee" page AND the login details of a new franchisee are entered incorrectly (email is already recorded in the system) **When** the user clicks on the "Add" button **Then** a message is displayed "Franchisee is NOT added to the system. This email is already recorded in the system."		

Let's discuss this User Story Card:
Are the ACs correct and complete?

Story # 5		Priority	Medium
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How easy would be to specify these acceptance criteria in the *Verify-That* format?

Story # 5		Priority	Medium
		Effort	8
As	An admin		
I want to	to delete a hospital staff member		
So that	they are no longer in the system		
Acceptance criteria	**AC 5.1** **Given** I am logged in as an admin **When** I click on “Management” tab **Then** I am forwarded to the page “Staff Management” **AC 5.2** **Given** I am logged in as an admin AND I am on the page “Staff Management” **When** I click on “Delete” button near the name of a staff member **Then** a pop-up window is displayed “Do you want to remove the staff member <name, ID> from the system?” with the buttons “Confirm” and “Cancel” **AC 5.3** **Given** I am logged in as an admin AND a pop-up window is displayed “Do you want to remove the staff member <name, ID> from the system?” **When** I click “Confirm” button **Then** the user <name, ID> is removed from the system and you get a confirmation message on the screen “The staff member <name, ID> removed from the system” **AC 5.4** **Given** I am logged in as an admin AND a pop-up window is displayed “Do you want to remove the staff member <name, ID> from the system?” **When** I click “Cancel” button **Then** the user <name, ID> is not removed from the system and you get a confirmation message on the screen “you cancelled the request”		

Are the ACs correct and complete?
Any other issues?

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You would need to check whether PBI ID, priority and efforts match to the information in the product backlog

You would need to check whether ACs’ IDs match to the PBI ID

I-want-to part should be reformulated to “delete a record of a hospital staff member” or “delete a hospital staff member’s record”

Story # 5		Priority	Medium
		Effort	8
As	An admin		
I want to	view status history of patient		
So that			
Acceptance criteria	**AC 5.1** **Given** I am logged in as an admin **When** I click on “Patients” tab **Then** I am forwarded to the page “Patient Management” **AC 5.2** **Given** I am logged in as an admin AND I am on the page “Patient Management” **When** I click on “History” button near the name of a patient **Then** the status history of the patient is presented in a new tab of browser”		

Are the ACs correct and complete?
Any other issues?

Story # 5	Priority	Medium
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As	An admin	
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So-that part is missing

You need to find the patient first…

Story # 5	Priority	Medium
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So that		
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You need to find the patient first…

AC 11.1
Given I am logged in as an admin AND I am on the “Search page” with a list of patients presented based on my search criteria
When I click on “History” button near the name of a patient
Then the status history of the patient is presented in a new tab of browser

Story # 8		Priority	Medium
		Effort	8
As	a doctor/nurse		
I want to	edit patient information		
So that	the patient record is up to date		
Acceptance criteria	**AC 8.1** **Given** I am logged in as a doctor/nurse **When** I click on “edit” button **Then** I can edit patient record		

Are the ACs correct and complete?
Any other issues?

Story # 8		Priority	Medium
		Effort	8
As	a doctor/nurse		
I want to	edit patient information		
So that	the patient record is up to date		
Acceptance criteria	**AC 8.1** **Given** I am logged in as a doctor/nurse **When** I click on “edit” button **Then** I can edit patient record		

AC 8.1-8.3… Do we need more?

AC 8.1

Given I am logged in as a doctor/ nurse AND I am on the “Search page” with a list of patients presented based on my search criteria

When I click on “Edit” button near the name of a patient

Then a new page “Patient Data” is opened with editable fields on patient information (full name, phone number, postal address) AND “Cancel” button is enabled AND “Save” button is disabled

AC 8.2

Given I am logged in as a doctor/ nurse AND I am on the page “Patient Data”

When I click on “Cancel” button

Then the page “Patient Data” will be closed AND no changes in patient’s data will be made

AC 8.3

Given I am logged in as a doctor/ nurse AND I am on the page “Patient Data” AND “Save” button is enabled

When I click on “Save” button

Then the page “Patient Data” will be closed AND all changes in patient’s data will be saved in the system

AC 8.4-8.6… Do we need more?

AC 8.4

Given I am logged in as a doctor/ nurse AND I am on the page “Patient Data”

When I edited the first name of the patient as per system constraints (the field contains only letters and is nonempty)

Then “Save” button is enabled

AC 8.5

Given I am logged in as a doctor/ nurse AND I am on the page “Patient Data”

When I incorrectly edited the first name of the patient (the field now contains not only letters or is empty)

Then “Save” button is disabled

AC 8.6

Given I am logged in as a doctor/ nurse AND I am on the page “Patient Data”

When I edited the last name of the patient as per system constraints (the field contains only letters and is nonempty)

Then “Save” button is enabled

AC 8.7… Do we need more?

AC 8.7

Given I am logged in as a doctor/ nurse AND I am on the page “Patient Data”

When I incorrectly edited the last name of the patient (the field now contains not only letters or is empty)

Then “Save” button is disabled